8 September, 2025

Bangladesh Army University of Science and Technology (BAUST), Saidpur

B.Sc. Engineering in Civil Engineering (CE)

List of Courses (Syllabus 2022, Effective from Batch 12 – Onwards)

Arts & Science Department		Cr.
HUM 1109	Professional English	2.00
HUM 1110	Technical Report Writing and Presentation	1.50
HUM 1209	Sociology	2.00
HUM 1219	Bengali Language and Literature	2.00
HUM 2109	Engineering Economics	2.00
HUM 2209	Financial and Managerial Accounting	2.00
HUM 2229	History of the Emergence of Bangladesh	2.00
PHY 1109	Physics I	3.00
PHY 1209	Physics II	3.00
PHY 1210	Physics Sessional	1.50
CHEM 1109	Chemistry	3.00
CHEM 1110	Chemistry Sessional	1.50
MATH 1109	Math I (Differential Calculus, Integral Calculus and Coordinate Geometry)	3.00
MATH 1209	Math II (Ordinary and Partial Differential Equations and Matrices)	3.00
MATH 2109	Math III (Vector Analysis, Laplace Transform and Statistics)	3.00
	Arts & Science Total Credits	**34.50**

Inter-Disciplinary Department		
Department of EEE		
EEE 1213	Basic Electrical Engineering	3.00
EEE 1214	Basic Electrical Engineering Sessional	1.50
Department of CSE		
CSE 2123	Introduction to Computer Programming	3.00
CSE 2124	Introduction to Computer Programming Sessional	1.50
Department of ME		
ME 1112	Mechanical Engineering Shop Practice	1.50
	Inter-Disciplinary Department Total Credits	**10.50**

Elective Courses		
Option I	***Civil Engineering Practice***	
CE 4203	Sustainability of Development Projects	2.00
CE 4205	Business and Career Development	2.00
Option II	***Structural Engineering***	
CE 4211	Dynamics of Structures	2.00
CE 4213	Prestressed Concrete	2.00
CE 4215	Steel Structures	2.00
CE 4217	Introduction to Finite Element Methods	2.00
CE 4210	Structural Analysis and Design Sessional II	1.50
CE 4220	Computer Aided Analysis and Design of Structures Sessional	1.50
Option III	***Geotechnical Engineering***	
CE 4231	Earth Retaining Structures and Machine Foundation	2.00
CE 4233	Soil Water Interaction	2.00
CE 4235	Soil Structure Interaction	2.00
CE 4230	Geotechnical Engineering Sessional II	1.50
Option IV	***Environmental Engineering***	
CE 4241	Solid and Hazardous Waste Management	2.00
CE 4243	Environmental Pollution Control	2.00
CE 4245	Environmental Development Project	2.00
CE 4240	Environmental Engineering Sessional II	1.50
Option V	***Transportation Engineering***	
CE 4251	Traffic Engineering Design and Management	2.00
CE 4253	Pavement Management, Drainage and Airport	2.00
CE 4255	Transportation Project and Management	2.00
CE 4250	Transportation Engineering Sessional II	1.50
Option VI	***Water Resources Engineering***	
CE 4261	Flood Control and Management	2.00
CE 4263	River Engineering	2.00
CE 4265	Coastal Engineering	2.00
CE 4267	Ground Water Engineering	2.00
CE 4269	Hydraulic Structures	2.00
CE 4260	Hydrology and Irrigation Engineering Sessional	1.50
	Elective Courses Total Credits to be taken	**11.00**
	TOTAL CREDITS	**160.00**

Core Courses of CE Program		
CE 1100	Civil Engineering Drawing	1.50
CE 1103	Surveying	3.00
CE 1200	Computer Aided Drawing	1.50
CE 1204	Practical Surveying	1.50
CE 1211	Engineering Mechanics	3.00
CE 2100	Details of Construction	1.50
CE 2105	Engineering Materials	3.00
CE 2106	Engineering Materials Sessional	1.50
CE 2111	Mechanics of Solids I	3.00
CE 2204	Quantity Surveying	1.50
CE 2207	Applied Mathematics for Engineers	3.00
CE 2211	Mechanics of Solids II	3.00
CE 2212	Mechanics of Solids Sessional	1.50
CE 2231	Engineering Geology and Geomorphology	3.00
CE 2261	Mechanics of Fluids	3.00
CE 2262	Mechanics of Fluids Sessional	1.50
CE 3107	Numerical Methods for Engineers	2.00
CE 3108	Engineering Computation Sessional	1.50
CE 3111	Analysis and Design of Determinate Structures	3.00
CE 3113	Reinforced Concrete Structures I	3.00
CE 3131	Soil Mechanics	3.00
CE 3132	Geotechnical Engineering Sessional I	1.50
CE 3141	Environmental and Water Supply Engineering	3.00
CE 3142	Environmental Engineering Sessional I	1.50
CE 3211	Analysis of Lateral Loads and Deflection of Structures	3.00
CE 3212	Structural Analysis and Design Sessional I	1.50
CE 3213	Reinforced Concrete Structures II	3.00
CE 3214	Bridge Design Sessional	1.50
CE 3231	Foundation Engineering	3.00
CE 3251	Fundamentals of Transportation Engineering	3.00
CE 3252	Transportation Engineering Sessional I	1.50
CE 3261	Open Channel Flow	3.00
CE 3262	Open Channel Flow Sessional	1.50
CE 4000	Thesis	3.50
CE 4002	Capstone Project	3.00
CE 4101	Engineering Ethics, Professional Practices and Communication	3.00
CE 4102	Engineering Ethics, Professional Practices and Communication Sessional	1.50
CE 4111	Analysis and Design of Indeterminate Structures	3.00
CE 4112	Building Design Sessional	1.50
CE 4141	Wastewater Engineering and Sanitation	3.00
CE 4151	Pavement Design and Railway Engineering	3.00
CE 4161	Hydrology and Irrigation Engineering	3.00
CE 4170	Industrial Attachment	1.00
CE 4201	Project Planning and Construction Management	3.00
	Core Course Total Credits	**104.00**

Note 1:

i. CE 4000 and CE 4002 will be conducted throughout the span of 4th year (4-I and 4-II)

ii. CE 4170 will commence after the end of 3-II final exam for 2-3 weeks at convenient time.

Note 2:

i. Students will take 1 (one) thesis related optional theory course and corresponding 1 (one) sessional course from any elective course groups from option II to VI

ii. Students will also take 3 (three) more optional theory courses from other elective course groups from option I to VI, but not more than 1 (one) from any group and 1 (one) more sessional course from any corresponding groups.

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B.Sc. Engineering in Civil Engineering (CE)

Level and Term Wise Courses (Syllabus 2022, Effective from Batch 12 – Onwards)

Level 1, Term I		Cr.
CE 1103	Surveying	3.00
PHY 1109	Physics I	3.00
CHEM 1109	Chemistry	3.00
MATH 1109	Math I (Differential Calculus, Integral Calculus and Coordinate Geometry)	3.00
HUM 1109	Professional English	2.00
CE 1100	Civil Engineering Drawing	1.50
CHEM 1110	Chemistry Sessional	1.50
HUM 1110	Technical Report Writing and Presentation	1.50
ME 1112	Mechanical Engineering Shop Practice	1.50
	Semester Total Credits	**20.00**

Level 1, Term II		Cr.
CE 1211	Engineering Mechanics	3.00
PHY 1209	Physics II	3.00
MATH 1209	Math II (Ordinary and Partial Differential Equations and Matrices)	3.00
HUM 1209	Sociology	2.00
EEE 1213	Basic Electrical Engineering	3.00
HUM 1219	Bengali Language and Literature	2.00
CE 1200	Computer Aided Drawing	1.50
CE 1204	Practical Surveying	1.50
PHY 1210	Physics Sessional	1.50
EEE 1214	Basic Electrical Engineering Sessional	1.50
	Semester Total Credits	**22.00**

Level 2, Term I		Cr.
CE 2105	Engineering Materials	3.00
CE 2111	Mechanics of Solids I	3.00
MATH 2109	Math III (Vector Analysis, Laplace Transform and Statistics)	3.00
HUM 2109	Engineering Economics	2.00
CSE 2123	Introduction to Computer Programming	3.00
CE 2100	Details of Construction	1.50
CE 2106	Engineering Materials Sessional	1.50
CSE 2124	Introduction to Computer Programming Sessional	1.50
	Semester Total Credits	**18.50**

Level 2, Term II		Cr.
CE 2207	Applied Mathematics for Engineers	3.00
CE 2211	Mechanics of Solids II	3.00
CE 2231	Engineering Geology and Geomorphology	3.00
CE 2261	Mechanics of Fluids	3.00
HUM 2209	Financial and Managerial Accounting	2.00
HUM 2229	History of the Emergence of Bangladesh	2.00
CE 2204	Quantity Surveying	1.50
CE 2212	Mechanics of Solids Sessional	1.50
CE 2262	Mechanics of Fluids Sessional	1.50
	Semester Total Credits	**20.50**

Level 3, Term I		Cr.
CE 3107	Numerical Methods for Engineers	2.00
CE 3111	Analysis and Design of Determinate Structures	3.00
CE 3113	Reinforced Concrete Structures I	3.00
CE 3131	Soil Mechanics	3.00
CE 3141	Environmental and Water Supply Engineering	3.00
CE 3108	Engineering Computation Sessional	1.50
CE 3132	Geotechnical Engineering Sessional I	1.50
CE 3142	Environmental Engineering Sessional I	1.50
	Semester Total Credits	**18.50**

Level 3, Term II		Cr.
CE 3211	Analysis of Lateral Loads and Deflection of Structures	3.00
CE 3213	Reinforced Concrete Structures II	3.00
CE 3231	Foundation Engineering	3.00
CE 3251	Fundamentals of Transportation Engineering	3.00
CE 3261	Open Channel Flow	3.00
CE 3212	Structural Analysis and Design Sessional I	1.50
CE 3214	Bridge Design Sessional	1.50
CE 3252	Transportation Engineering Sessional I	1.50
CE 3262	Open Channel Flow Sessional	1.50
	Semester Total Credits	**21.00**

Level 4, Term I		Cr.
CE 4101	Engineering Ethics, Professional Practices and Communication	3.00
CE 4111	Analysis and Design of Indeterminate Structures	3.00
CE 4141	Wastewater Engineering and Sanitation	3.00
CE 4151	Pavement Design and Railway Engineering	3.00
CE 4161	Hydrology and Irrigation Engineering	3.00
CE 4000	Thesis	1.50
CE 4002	Capstone Project	1.50
CE 4102	Engineering Ethics, Professional Practices and Communication Sessional	1.50
CE 4112	Building Design Sessional	1.50
CE 4170	Industrial Attachment	1.00
	Semester Total Credits	**22.00**

Level 4, Term II		Cr.
CE 4201	Project Planning and Construction Management	3.00
CE 42XX	Elective theory course from Option I ~ VI	2.00
CE 42XX	Elective theory course from Option I ~ VI	2.00
CE 42XX	Elective theory course from Option I ~ VI	2.00
CE 42XX	Elective theory course from Option II ~ VI (Thesis related)	2.00
CE 4000	Thesis	2.00
CE 4002	Capstone Project	1.50
CE 42XX	Elective sessional course from Option II ~ VI (Thesis related)	1.50
CE 42XX	Elective sessional course from Option II ~ VI	1.50
	Semester Total Credits	**17.50**

ID:

Student Name:

Contact No: